

Article

Capsaicin-Modified Fluorosilicone Based Acrylate Coating for Marine Anti-Biofouling

Jie Liu ^{1,2,3}, Jiawen Sun ^{2,3}, Jizhou Duan ^{2,3,*} , Xucheng Dong ^{2,3}, Xinping Wang ^{1,*}, Chao Liu ⁴ and Baorong Hou ^{2,3}

¹ College of Chemistry and Molecular Engineering, Qingdao University of Science and Technology, 53 Zhengzhou Road, Qingdao 266042, China; a502075459@163.com

² Key Laboratory of Marine Environmental Corrosion and Bio-Fouling, Institute of Oceanology, Chinese Academy of Sciences, 7 Nanhai Road, Qingdao 266071, China; jiawensun1021@126.com (J.S.); dongxucheng@qdio.ac.cn (X.D.); baoronghou@163.com (B.H.)

³ Open Studio for Marine Corrosion and Protection, Pilot National Laboratory for Marine Science and Technology (Qingdao), 1 Wenhai Middle Road, Qingdao 266237, China

⁴ College of Resources and Environment, Hunan Agricultural University, 1 Nongda Road, Changsha 410128, China; bbhliu@126.com

* Correspondence: duanjz@qdio.ac.cn (J.D.); xpwang@qust.edu.cn (X.W.)

Abstract: Capsaicin has been extensively studied for its excellent antifouling activity and very low environmental toxicity. However, mixing natural capsaicin with coatings can cause rapid capsaicin leakage, severely shortening its antifouling cycle. In this study, we describe the preparation and performance of a new capsaicin-modified marine antifouling organofluorosilicone, which is based on silicone and fluorine acrylate monomers covalently bound to an organic antimicrobial monomer, HMBA (N-(4-hydroxy-3-methoxybenzyl)-acrylamide) on a polymer network. The chemical grafting of HMBA into the polymer has improved the problem of short antifouling life of the coating due to antifouling agent leakage and the environmental pollution caused by the leakage. The study focused on the synthesis of pristine acrylate monomers with organic bioactive groups prepared from vanillin amine salts and their co-polymerization in the presence of distal acrylate oligomers. The resulting cross-linked films were characterized using infrared spectroscopy, contact angle, and adhesion analyses. The results indicate that the materials had good adhesion, low surface energy, and were resistant to prolonged immersion in water. The polyacrylate coating synthesized from acrylate exhibited antibacterial and anti-algae activity. Biological tests on the marine microorganisms, *Pseudomonas species*, *Shewanella species*, and *Navicula incerta*, revealed a 97%, 98%, and 99% reduction compared to the blank control group, respectively, indicating that the coating has strong anti-adhesive ability. This work is expected to develop a promising material for marine antifouling.

Keywords: capsaicin; fluorosilicone; acrylic; low surface energy; marine anti-biofouling



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1. Introduction

Marine engineering and ship surfaces exposed to seawater are colonized by bacteria, algae and other macrofouling organisms, leading to higher fuel consumption, reduced speed, increased greenhouse emissions, and clogging of seawater drainage pipes [1–3]. Currently, the use of antifouling coatings containing biocides is the most effective method of reducing the adhesion of fouling organisms. This mainly relies on self-polishing coatings that release toxic molecules at a controlled rate to suppress the growth of bacteria, algae, and macroscopic fouling organisms. However, these biocides have low specificity and their off-target effects negatively impact the environment [4,5]. Thus, less toxic antifouling coatings that are environmentally friendly are urgently needed. Natural products are a promising source of biodegradable, broad-spectrum, antifouling agents with low environmental pollution when compared to heavy metals. Such natural products are highly attractive due to their

capacity to inhibit the adhesion of microorganisms without killing them [6–9]. Capsaicin (8-methyl-N-vanillyl-6-nonenamide), the main active ingredient of *Capsicum annum*, has been shown to effectively suppress the adhesion of marine organisms [8]. Its excellent antifouling properties and remarkably low environmental toxicity have been extensively studied since the first report on capsaicin-containing antifouling coatings in 1995 [2,10]. Organic coatings are currently an area of intense research for the creation of formulations that contain capsaicin for improved antifouling activity. For example, a capsaicin-modified antimicrobial acrylate polymer with high antimicrobial activity has been developed. The incorporation of capsaicin in silicone coatings has also been studied. However, natural capsaicinoids are usually mixed with coatings, which causes rapid capsaicinoid leakage, severely shortening their antifouling cycle [11]. To overcome this challenge, the chemical synthesis of these compounds has been proposed. Yu et al. have synthesized various modified substances of capsaicin and found that they all have antibacterial activity [12–15]. Zhang et al. [16] developed an “in situ polymerization-mixing” technique for controlling capsaicin leaching, which can maintain good stability for over 2 months. Here, we synthesized a modified monomer of capsaicin, N-(4-hydroxy-3-methoxybenzyl)-acrylamide (HMBA) and copolymerized it with silicone- and organofluorine-modified acrylate to make a hydrophobic coating [17,18]. Acrylates are widely used in coatings due to their excellent film-forming properties and high adhesion to substrates. Silicone-modified acrylates have good mechanical properties, are resistant to heat and adverse weather conditions, and produce films with low surface energy [19–21]. After curing, the fluorine-containing side chains stretch at the interface, significantly reducing the film and conferring the polymer with strong water and oil repellency, improving the friction and corrosion resistance of the film [22–24].

In this study, HMBA-containing polyacrylic resin coating was prepared using free radical copolymerization reaction to achieve the chemical grafting of capsaicin in polymers. The structural, surface, and mechanical properties of the polymer were characterized using IR and NMR hydrogen spectroscopy, as well as surface energy, adhesion, and hardness tests. The HMBA was homogeneously dispersed in the polymer using EDS analysis. To determine the membranes’ antifouling properties, we evaluated the ability of bacteria and algae to colonize the membranes. Antifouling analysis showed that HMBA-containing coatings had good anti-adhesive properties against *Pseudomonas species*, *Shewanella species*, and *Navicula incerta* (*N. incerta*). Our objective was to investigate the feasibility of this technology to develop an environmentally friendly polyacrylate coating for marine antifouling.

2. Experimental

2.1. Materials

2,2,2-Trifluoroethyl methacrylate (98%), 1,6-Hexanediol diacrylate (>90%), 3-[Tris(trimethylsiloxy)silyl] propyl methacrylate (97%), Polydimethylsiloxane, Methylldintri-p-phenylene trilsocyanate (20% in C₆H₅Cl), 2,2'-azobis(2-methylpropionitrile) (AIBN, 99%), Acryloyl chloride (>96%) and Vanillylamine hydrochloride (99%) were purchased from Shanghai Aladdin Biochemical Technology Co. Ltd. (Shanghai, China). AIBN was recrystallized from ethanol. Tetrahydrofuran (THF) was purchased from Sinopharm Chemical Reagent (Tianjin, China).

2.2. Preparation of Capsaicin Modified HMBA

Vanillin hydrochloride (2 g, 10.6 mmol) was dissolved in 200 mL of deionized water and 200 mL of 0.5 N/L sodium hydroxide added. The solution was then stirred and filtered to get white crystals, which were then vacuum dried overnight at room temperature. A total of 500 mg (6.5 mmol) of purified vanillin was put in a reaction flask and dissolved in 50 mL tetrahydrofuran. Next, 0.9 mL (6.45 mmol) of triethylamine and 235 μ L (2.95 mmol) of acryloyl chloride were added on an ice bath and the solution was stirred for 3 h at room temperature under N₂ protection. The mixture was then extracted with CH₂Cl₂ and the organic phase washed with 0.01 mol/L hydrochloric acid and then with saturated salt

water. Next, it was dried using anhydrous magnesium sulfate with spinning to obtain HMBA (N-(4-hydroxy-3-methoxybenzyl)-acrylamide), a light-yellow solid (Yield of 70%).

2.3. Preparation of PRE Coatings

1,6-hexanediyl diacrylate (1.41 g, 6.2 mmol), methacryloxypropyl (trimethylsiloxy) silane (2.64 g, 6.2 mmol), trifluoroethyl methacrylate (0.21 g, 1.25 mmol), and HMBA (0.43 g, 2.1 mmol) were dissolved in 50 mL anhydrous tetrahydrofuran with 1% equivalent of AIBN and mixed at 75 °C for 24 h. After the reaction, the product does not experience gelation and the solvent was spun at the end of the reaction, the solvent was spin dried to give a light-yellow viscous liquid, which was washed three times with methanol to remove the unreacted monomers. Next, 4 g of polymer was taken, 2 mL of tetrahydrofuran was added, the solution was mixed well, and then triphenyl triisocyanate (0.77 g, 2.1 mmol) was added. The coating was cured at 80 °C for 4 h. The membrane was recorded as PRE-10. The amount of HMBA was then changed to 0 g (0 mmol), 0.86 g (4.2 mmol) and 1.29 g (6.2 mmol), and the other substances were kept constant to produce the coatings as PRE-0, PRE-20 and PRE-30, respectively.

2.4. Characterization

A Nicolet IS5 Fourier transform infrared spectrometer (ThermoScientific, Waltham, MA, USA) was used for Fourier transform infrared spectroscopy using the KBr compression method and a KBr to sample mass ratio of 100:1. Each spectrum was based on 32 scans between 4000 and 500 cm^{-1} . NMR analysis was done on an AV400 MHz NMR spectrometer (Bruker, Karlsruhe, Germany), with DMSO as test reagent and TMS (0.03% V/V) as internal standard at 16 testing cycles for each sample. THF gel permeation chromatography (GPC) was performed in tetrahydrofuran at 35 °C at a flow rate of 1 mL/min on an Agilent 1260 HPLC system equipped with a G7110B pump and a G7162A refractive index detector. Number-average (M_n) and heavy-average molecular weights (M_w) of the polymers, as well as their ratios (polydispersity index, $\text{PDI} = M_w/M_n$) were estimated.

2.5. Water Contact Angles and Surface Free Energy

Contact angles and surface free energy were determined using a contact angle tester (OCA 25, Dataphysics, Hamburg, Germany) by placing 3 μL of a solidifying solution on the coating surface at room temperature. Static water contact angle was measured at 5 different areas of each sample and averaged. Contact angle was measured in the same way as the aqueous phase using diiodomethane as oil phase. Surface free energy was calculated using the Owens-Wendt-Rabel-Kaelble method based on the measured contact angles of deionized water (DI) and diiodomethane (DIM).

2.6. Adhesion Test

Adhesion between the sample with steel and the glass fiber reinforced epoxy substrate was measured using Defelsko's PosiTest AT-A pull-off adhesion tester (New York, NY, USA) according to ASTM D4541. Adhesion force was measured by a pull-off test in which aluminum spindles were separated from the substrate at 5 different locations at 0.2 MPa s^{-1} , and the average adhesion force calculated.

2.7. Pencil Hardness Test

The "Pencil hardness of coating film" for rough hardness testing of the prepared coating was determined according to GB/T 6739-1996. The tester for determining the pencil hardness of the coating film consists of a pencil jig and moving table for the coating film sample. The sample film to be tested was fixed face up on the moving table, and the pencil clamped on the pencil jig at a 45° angle to the plane of the coating film ensuring that the pencil tip was pressed tightly against the coating film. The film was then scratched and examined for obvious pencil scratches on the surface of the coating. The testing was

done from high to low hardness in turn, until no scratches were observed. Pencil hardness ranged from 9H to 6B.

2.8. Antibacterial Assays

Next, we used *Pseudomonas* and *Shewanella species*, which are commonly used to evaluate the performance of antimicrobial materials, to test the antimicrobial activity of the coatings. *Pseudomonas* and *Shewanella species* were cultured in liquid LB media (10 g Tryptone, 10 g NaCl, 5 g Yeast extract and 1 L deionized water) at 37 °C for 10 h and bacteria concentration determined in colony forming units (CFU). Next, the bacteria were pelleted from the media by centrifugation at $1324 \times g$ for 5 min. The bacteria pellets were then resuspended in 0.1 M PBS (8.0 g NaCl, 0.2 g KCl, 1.44 g Na_2HPO_4 and 0.44 g KH_2PO_4 and 1 L deionized water) at a density of 10^7 CFU ml^{-1} for use in antimicrobial assays. Control glass slides (uncoated) and glass slides coated with PDMS elastomer, PRE-0, PRE-10, PRE-20, or PRE-30 were cut into 1 cm $\times$ 1 cm pieces and sterilized under UV light for 30 min. They were then placed in the bacterial suspension for 6 h. Next, the slides were subjected to live/dead staining using a BacLight bacterial viability kit for 15 min. Excess stain and poorly adherent bacteria were then washed off using PBS and adherent bacteria examined under a fluorescent microscope (BX-51, Olympus, Tokyo, Japan). Take the same three replicate samples for each material, and the bacteria counted in five fields of view per sample. The bacteria anti-adhesion rate (R) was calculated using the formula:

$$R = (N_c - N_s) / N_c \quad (1)$$

where N_c and N_s are the number of bacteria observed on the uncoated and coated glass slides, respectively. Bacteria media had been autoclaved before use and the inoculation and assembly steps were done on an ultra-clean bench after UV sterilization for 30 min.

2.9. Diatom Settlement

The diatom *N. incerta* was obtained from the State Key Laboratory of Marine Corrosion and Protection, Qingdao. Diatom cell sedimentation experiments were done on uncoated glass slides (control) and glass slides coated with PDMS elastomer, PRE-0, PRE-10, PRE-20, or PRE-30. The samples were placed in six-well plates and 10 mL of diatom cell suspension (1.2×10^5 cells mL^{-1}) added into each well. Samples were then incubated at 23 °C for 21 days in 12-h light/dark cycles. Samples were picked on day 1, day 7, day 14 and day 21 and washed with sterilized seawater to remove poorly attached diatoms. They were then observed under a fluorescence microscope (BX-51, Olympus, Tokyo, Japan) with the same three replicate samples for each sample, and diatoms counted in 5 fields of view per sample.

3. Results and Discussion

3.1. Structural Characterization

HMBA is synthesized from vanillin ammonium salt and acryloyl chloride. The capsaicin derivatives, HMBA, 2,2,2-Trifluoroethyl methacrylate, and 1,6-Hexanediol diacrylate 3-[Tris(trimethylsiloxy)silyl] propyl was synthesized using commercially available vanillin amine hydrochloride. Acryloyl chloride methacrylate and Methylldintri-p-phenylen trisocyanate were subjected to a two-step reaction to synthesize the polymeric membrane PRE. The synthesis process is shown in Figure 1.

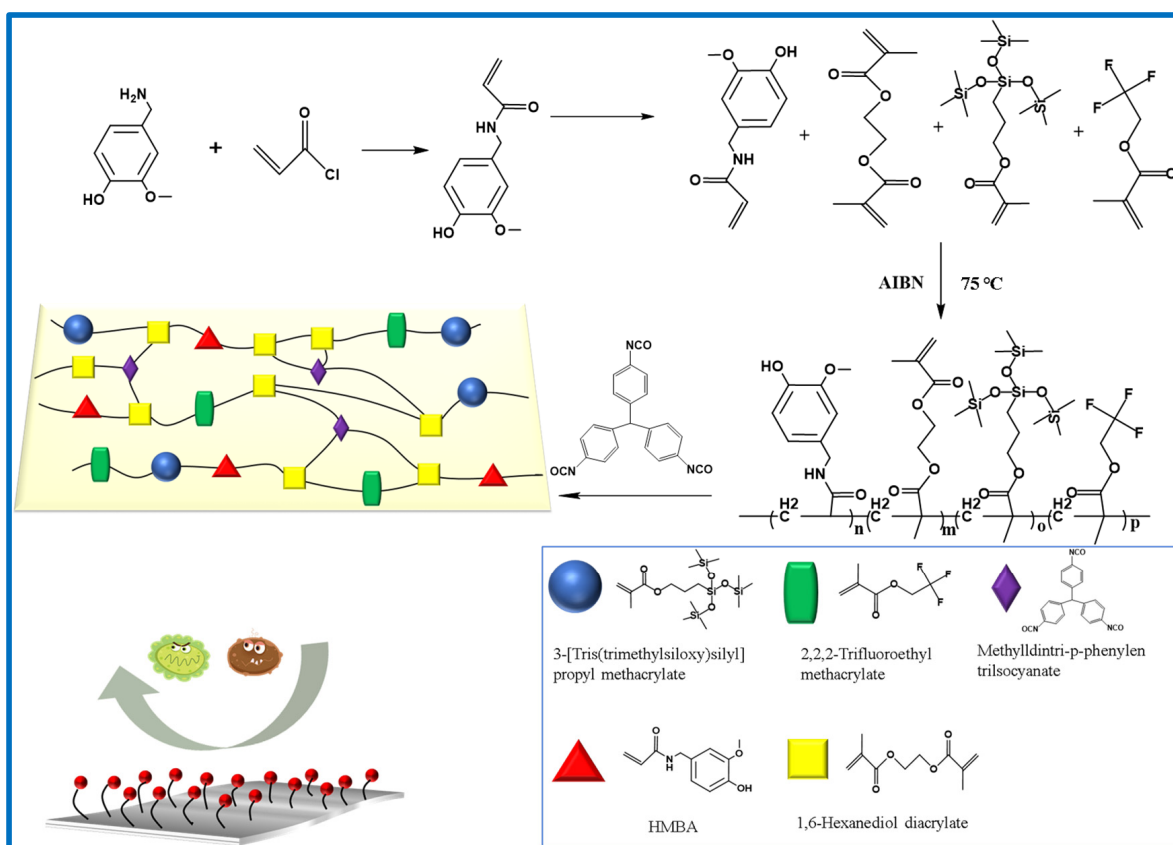


Figure 1. A schematic representation of the preparation of the PRE coating.

Figure 2A,B show the HMBA's infrared and NMR hydrogen spectra, respectively. In the infrared spectrum of HMBA, the absorption peaks at 3324 and 3099 cm^{-1} belong to hydroxyl (-OH) and amino (-NH) groups. Peaks 1656 and 1597 cm^{-1} are the contraction vibration peaks of carbonyl (C=O) and double bond (C=C). The NMR hydrogen spectrum of 8.96 ppm corresponds to the proton signal on NH, which indicates successful HMBA synthesis.

The chemical structures of PRE-10, PRE-20 and PRE-30 were determined using ^1H NMR (Figure 2C). A benzene ring was strongly suggested by the signals at 5.8–6.3 ppm (C). The protons of Ph-OH and are represented by -NH 6.9 ppm (b) and 6.5 ppm (e). The peaks at 3.0 (a) and 5.5 (d) ppm belong to the protons on the Ph-OCH₃ and N-CH₂. The intensity of the above peaks increased significantly with HMBA content. The characteristic peaks at 0.1 to 0.3 ppm (g) correspond to Si-(CH₃)₃. 1.0–1.7 ppm (i, k, m, t, q, u, v, w, x, y) are the proton characteristic peaks of -CH₃ and -CH₂. Proton signals that appear between 1.8 and 2.0 ppm (f, s) represent O=C-CH₂. Characteristic peaks at around 3.5–3.7 ppm (j, r, p) correspond to the proton on the carbon attached to the ester group. The peaks at around 3.8–4.0 ppm (o, b) correspond to the double bond close to the carbonyl. The presence of protons attached to the CF₃ group 5.6 ppm (l) was also observed.

Figure 2D exhibits the typical FT-IR spectra obtained from PRE-0, PRE-10, PRE-20 and PRE-30. For PRE-10, PRE-20 and PRE-30 spectra, the weak absorption peak at 3320 cm^{-1} and the weak absorption peak at 1514 cm^{-1} contraction vibration belong to the amino and benzene groups. The absence of these two peaks in PRE-0 indicates HMBA polymerization into the polymer's main chain. The absorption peaks at 2955, 2870 and 1728 cm^{-1} are methyl (-CH₃), methylene (-CH₂) and carbonyl (C=O), respectively. The absorption peak of CF₃ and Si-O-Si is 1048 cm^{-1} . Peak 1255 cm^{-1} is the symmetric stretching vibration characteristic of C-O and the symmetric deformation absorption peak of Si-(CH₃)₃. Peak 1160 cm^{-1} is the antisymmetric stretching vibration characteristic of C-O. The weak peak at 1669 cm^{-1} is the stretching vibration peak of C=C conjugated to C=O.

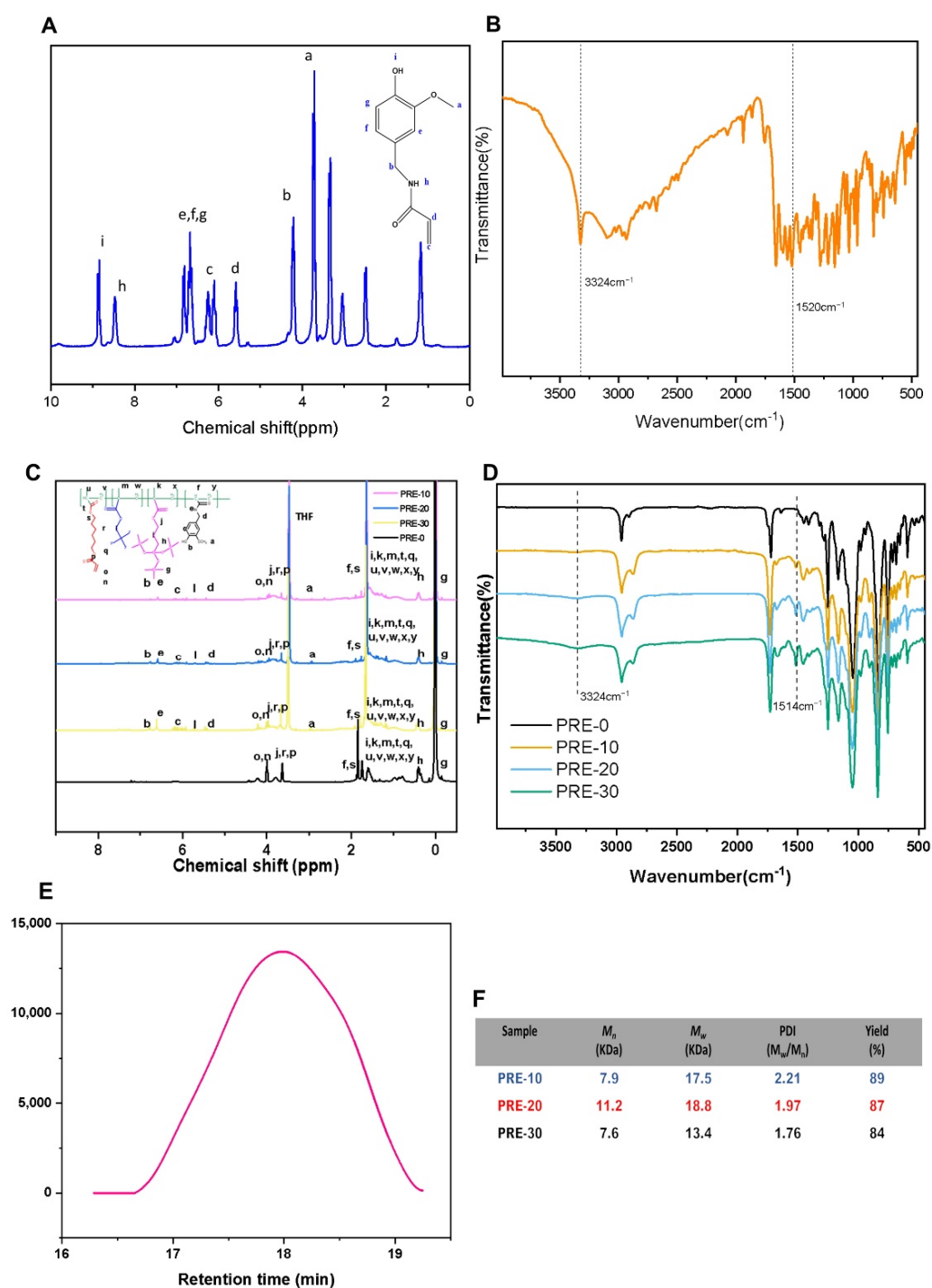


Figure 2. The structure of PRE polymer (A) ¹H NMR of HMBA, (B) FTIR spectrum of HMBA, (C) ¹H NMR of PRE polymer, (D) FTIR spectrum of PRE polymer, (E) Gel permeation chromatography of PRE-0, (F) Molecular weight and polydispersity and yield of the polymer.

Gel permeation chromatography (GPC) analysis revealed the molecular weight ($M_w = 13.4$ – 18.8 kDa) and narrow polydispersity ($PDI = 1.76$ – 2.21) of the PRE polymers, as well as their high yields (84–89%) (Figure 2F).

3.2. Mechanical Properties

It has been reported that the integrity and effective duration of polymer coating underwater depends largely on the magnitude of its adhesion to the substrate [25]. The adhesion of the coating depends on the thickness of the coating and the roughness of the

coating surface, among other factors. This has been mentioned in our previous studies [26]. The lowest adhesion strength of PRE-0 coating to glass and steel plates was 1.9 ± 0.05 and 2.1 ± 0.05 MPa, respectively, which is already a very favorable adhesion strength. For PRE-10, PRE-20 and PRE-30, 2.0 ± 0.02 , 2.1 ± 0.04 and 2.1 ± 0.05 MPa were obtained, respectively. It can be noticed that the adhesion strength of the hybrid coatings increased with the increase in HMBA dosage, which may be due to the increase in hydrophilic groups in the polymer leading to a gradual increase in the surface energy of the coatings. Similarly, the adhesion strengths of PRE-0, PRE-10, PRE-20 and PRE-30 on steel plates were 2.2 ± 0.02 , 2.3 ± 0.02 , 2.4 ± 0.04 and 2.45 ± 0.05 MPa, respectively, which showed a similar trend to the adhesion strength of the hybrid coatings on glass. In addition, the reason for the higher adhesion of hybrid coatings on steel surface than on glass is that the steel surface has more hydroxyl groups, which can form more hydrogen bonds with the hybrid coatings. More importantly, all the hybrid coatings still showed excellent adhesion performance after 30 days of immersion in ASW (Artificial seawater configured according to ASTM D1141-98 standard), similar to the values measured before immersion, proving that all the coatings have stable adhesion performance in the marine environment [27,28].

The hardness of the coating which was determined based on the pencil hardness is 2H. This level of hardness can effectively prevent mechanical damage and prolong antifouling process.

3.3. Surface Properties

The contact angles of water and diiodomethane for PRE-0, PRE-10, PRE-20, and PRE-30 coatings are illustrated in Figure 3B. As expected, the contact angle of water decreases when the amount of HMBA is increased from $105.6^\circ \pm 0.2^\circ$ to $98.5^\circ \pm 0.2^\circ$. Similarly, the contact angle of diiodomethane decreases from $68.9^\circ \pm 0.5^\circ$ to $64.8^\circ \pm 0.3^\circ$. The surface energies of PRE-0, PRE-10, PRE-20 and PRE-30 were 23.43 ± 0.12 , 23.53 ± 0.18 , 24.50 ± 0.25 and 26.03 ± 0.33 , respectively. The results show that PRE has the lowest surface energy, implying that it can more effectively resist adherence of fouling organisms [29]. In addition, contact angles were determined on the PRE samples after immersion in seawater daily for 30 days. The contact angle of water was stable for all samples and the samples showed stable surface properties. SEM analysis showed that the surfaces of all samples were smooth, and the EDS pattern (C: 46.9 Wt. % N: 9.5 Wt. % O: 28.5 Wt. % Si: 5.4 Wt. % F: 9.7 Wt. %) images show that the various polymers are uniformly dispersed, especially the N element in HMBA is uniformly distributed in the coating.

These mechanical and surface properties indicate that PRE coatings have stable surface wettability, low surface energy, smooth surfaces and strong adhesion ability to substrates. This implies that PRE coatings can improve resistance to fouling organisms and enhance durability of samples underwater.

3.4. Antibacterial and Anti-Diatom Properties of the Surface

Anti-bacterial adhesion and diatom deposition experiments were conducted to explore the antifouling effect of the coating. *Shewanella* and *Pseudomonas* bacterial species were selected as model bacteria to evaluate antibacterial properties of the coatings [30,31]. Fluorescence microscopy showed that the anti-adhesion rates of PRE-30, PRE-20, PRE-10, and PRE-0 compared with the control glass slides for *Shewanella species.* and *Pseudomonas species.* adhesion were $97.3 \pm 0.6\%$ and $98 \pm 0.4\%$, $93 \pm 1.1\%$ and $95 \pm 1.5\%$, $80 \pm 3.0\%$ and $82 \pm 2.0\%$, $73 \pm 3.6\%$ and $70 \pm 3\%$ (Figure 4A,B). Anti-adhesion rates were higher for PRE-0, PRE-10, PRE-20 and PRE-30 compared with that of PDMS ($50 \pm 1.3\%$ and $48 \pm 2\%$ for *Shewanella species.* and *Pseudomonas species.*, respectively). This observation is attributed to incorporation of HMBA and the lower surface energy of the coating act which synergistically resist bacterial adhesion [31].

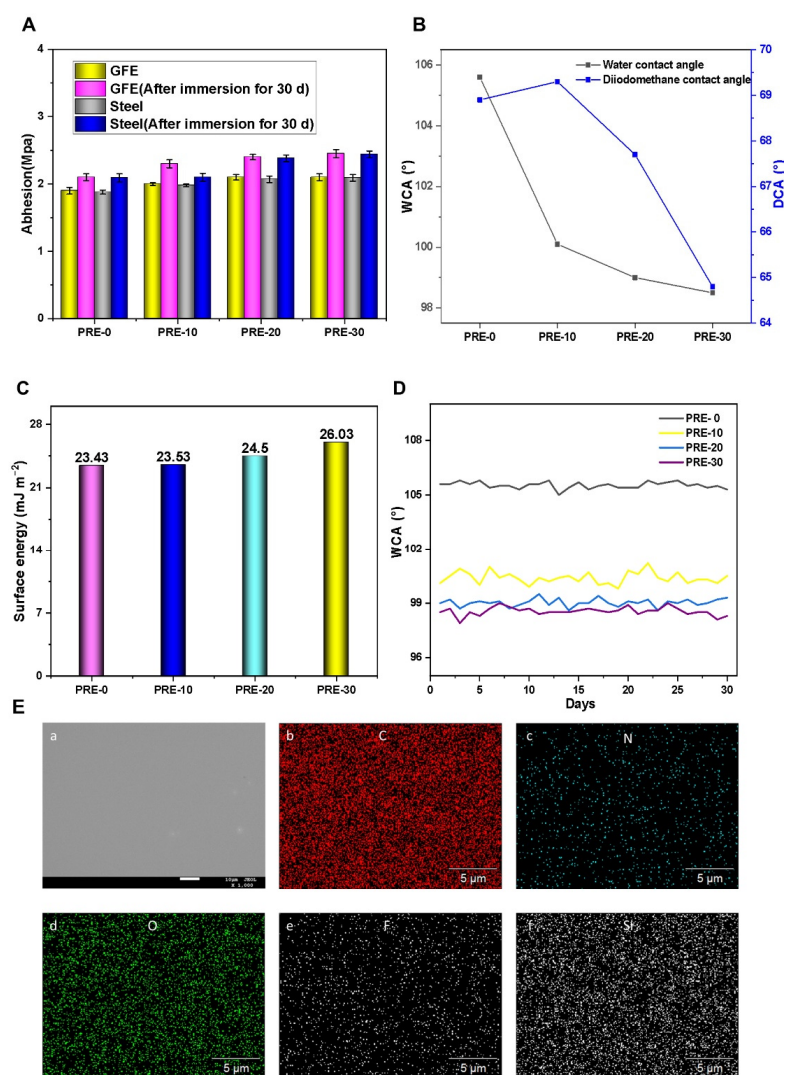


Figure 3. (A) Adhesion test results of coatings on Steel and GFE (Glass fiber reinforced epoxy resin) (B) WCA (Water contact angle) and DCA (Diiodomethane contact angle) for coating (C) Surface energy of the coating. (D) Contact angle of the coating immersed in water daily for 30 days. (E) (a) SEM analysis; (b–f) EDS mapping images of the PRE-20.

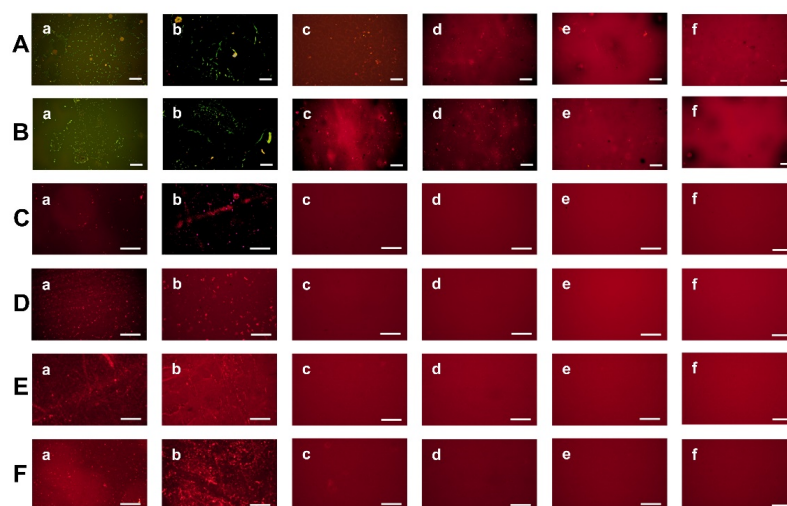


Figure 4. Cont.

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Sample	Adhesion rate(<i>P.sp.</i>)	Adhesion rate(<i>S.sp.</i>)
Glass	100%	100%
PDMS	52±5%	50±1.3%
PRE-0	30±3.6%	27±3%
PRE-10	18±2%	20±3%
PRE-20	5±1.5%	7±1.1%
PRE-30	2±0.4%	1.7±0.6%

H

Sample	Number of adhesions (cells mm ⁻²) (1d)	Number of adhesions (cells mm ⁻²) (7d)	Number of adhesions (cells mm ⁻²) (14d)	Number of adhesions (cells mm ⁻²) (21d)
Glass	45±5	192±20	448±25	585±50
PDMS	60±10	203±30	652±40	964±60
PRE-0	23±3	30±5	39±8	55±10
PRE-10	11±2	15±2	28±5	23±5
PRE-20	1±1	1±1	2±1	1±1
PRE-30	1±1	2±1	1±1	1±1

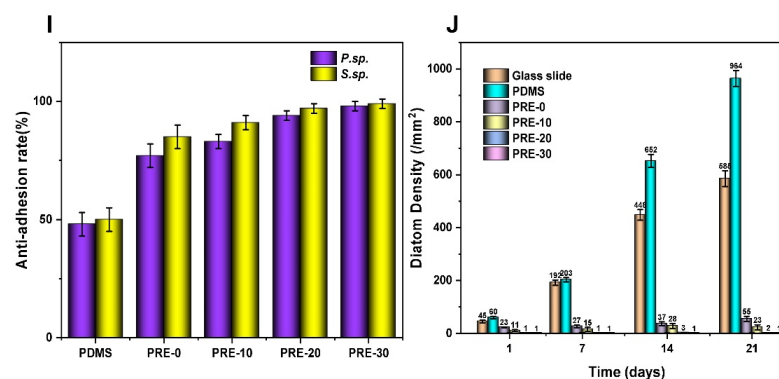


Figure 4. Antibacterial and anti-silica properties of PRE polymers. (A) Fluorescence images of *Pseudomonas sp.* (legend is 50 μ m). (B) Fluorescence images of *Shewanella sp.* (legend is 50 μ m). (C–F) Fluorescence images of *N. incerta* adhesion after 1, 7, 14 and 21 days to (a) glass, (b) PDMS, (c) PRE-0, (d) PRE-10, (e) PRE-20 and (f) PRE-30 fluorescence images (legend for 100 μ m). (G) Adhesion rates of *P. sp.* and *S. sp.* (H) The number of *N. incerta* settling on the coating surface at different days (I) Adhesion rates of *Pseudomonas sp.* and *Shewanella sp.* (J) Quantitative sedimentation density of *N. incerta* on the coating surface.

Further, the diatom resistance of the PRE coatings was evaluated using the diatom species, *Navicula incerta*. Fluorescence analysis of *Navicula incerta* adhered to the surface of each coating was performed and quantitative diatom sedimentation density after 1, 7, 14 and 21 days of immersion in diatom cell culture medium was determined (Figure 4C–F). Most *Navicula incerta* adhered to glass slides (1 day: ~45 cells mm⁻², 7 days: ~192 cells mm⁻², 14 days: ~448 cells mm⁻², 21 days: ~585 cells mm⁻²). Diatoms adhered more on PDMS (1: ~60 cells mm⁻², 7: ~203 cells mm⁻², 14: ~652 cells mm⁻², 21: ~964 cells mm⁻²) compared with the PRE coatings because they prefer hydrophobic surfaces. Less diatoms adhered to PRE-0 compared with those that adhered to PDMS (1 day ~23 cells mm⁻², 7 days, 14 days ~37 cells mm⁻², 21 days ~55 cells mm⁻²). This can be attributed to the low hydrophilic ability of PRE-0 compared with PDMS. Notably, there was little or no diatom adhesion on the surface of PRE-10, PRE-20 and PRE-30 after addition of HMBA (PRE-10: 1 day ~11 cells mm⁻², 7 days ~15 cells mm⁻², 14 days ~28 cells mm⁻², 21 days ~23 cells mm⁻²; PRE-20: 1 day ~1 cell mm⁻², 7 days ~1 cell mm⁻², 14 days ~3 cells mm⁻², 21 days ~2 cells mm⁻²; PRE-30: 1 day ~1 cell mm⁻², 7 days ~2 cells mm⁻², 14 days ~1 cell mm⁻², 21 days ~1 cell mm⁻²). This implies that

incorporation of HMBA effectively resisted adhesion of diatoms. The coating had similar anti diatom properties even after 1, 7, 14 and 21 days of immersion in diatom culture solution, which can be attributed to the permanent effect of HMBA on the surface of the coating. HMBA is a derivative of capsaicin. The findings in this study indicate that HMBA can act as a non-contaminating antifouling agent and can effectively prevent adhesion of bacteria and diatoms.

4. Conclusions

In summary, a capsaicin-modified fluorosilicone marine antifouling coating was synthesized in the present study by free radical copolymerization using three acrylates as raw materials. Spectral (FT-IR and $^1\text{H-NMR}$) analyses indicated successful synthesis of the PRE. The coating had excellent mechanical strength at was stable at ambient temperature. Biological tests showed that the prepared copolymer film had good antibacterial and anti-algal properties under static conditions. Copolymerization of capsaicin derivatives are effective long-term antifouling agents; therefore, the polymer is a promising material for marine antifouling.

Author Contributions: J.L.: Investigation, Formal analysis, Data curation, Visualization, Writing—original draft, J.S.: Formal analysis & Data curation, X.D.: Formal analysis & Data curation, X.W.: Methodology & Resources, B.H.: Supervision & Resources, C.L.: Formal analysis & Data curation J.D.: Conceptualization, Visualization, Methodology, Supervision, Resources. All authors have read and agreed to the published version of the manuscript.

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